香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Lecture 8: Advanced Concepts: Correlation and Conditional Probability

Shuang Li

Recap of Continuous Random Variable

Continuous R.V.

- A continuous random variable can take any value within its range (an interval of a union of multiple intervals of real numbers).

- We cannot list all the possible values and their probabilities as in the discrete case.

How to describe the probability?

- If P(E) = 0, then E is a zero-probability event.
- If E is empty, then E is impossible.

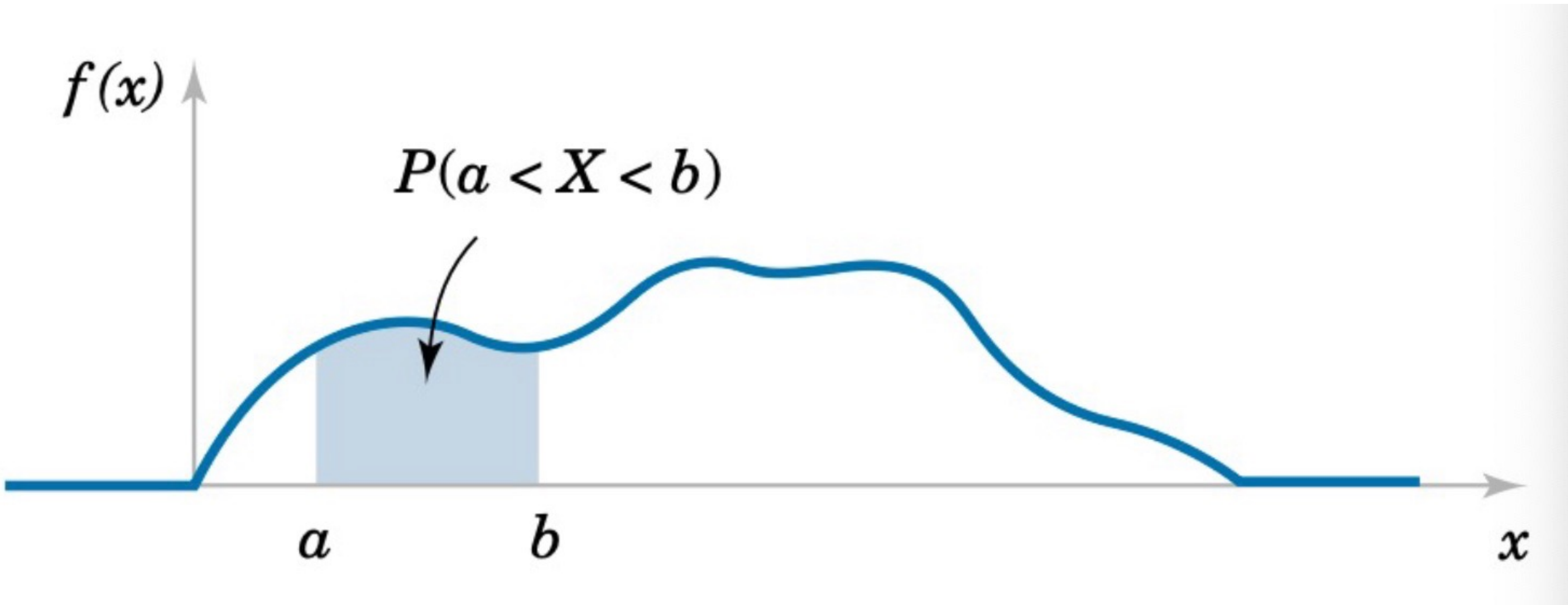
- For a continuous RV X, P(X=x) = 0 but {x} is not an impossible event.

- We will not use the probability mass function (pmf), namely P(X=x).

- Instead, we introduce a function f(ω), called the probability **density** function (pdf).
  - f(ω) > 0, if $\omega \in S$
  - f(ω) = 0, if $\omega \notin S$
  - $\int_{-\infty}^{\infty} f(x)dx = 1.$

Probability of $X \in [a, b]$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



Properties of PDF

- For x that is not in the sample space, $f(x)=0$

- A large value of $f(x)$ means that the values around x is more likely to be observed. (remember this implication)

- As a pdf, f(x) can be larger than 1, while as a pmf, f(x) cannot be larger than 1.
  - $f(\omega) = 2$, if $\omega \in [0, 0.5]$
  - $f(\omega) = 0$, if $\omega \notin [0, 0.5]$

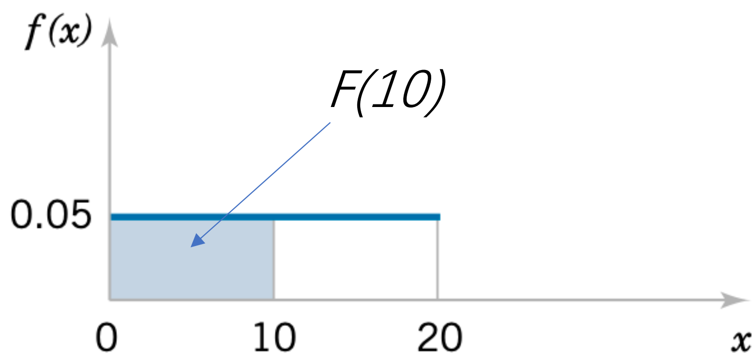
CDF

- Recall: the CDF of a discrete random variable X is

$$F(x) = P(X \leq x) = \sum_{\tilde{x} \leq x} f(\tilde{x})$$

- CDF for continuous random variable is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)du$$



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- CDF for continuous random variable is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)du$$

✓ $0 \leq F(x) \leq 1$
✓ If $x \leq y$, then $F(x) \leq F(y)$

For both discrete and continuous RVs

Mean and Variance

- Discrete:
  - ✓ Probability mass function.
- Continuous
  - ✓ Probability density function.

$\boxed{\textbf{Summation} \leftrightarrow \textbf{Integration}}$

- Mean

$$\mathrm{E}[X] = \sum x\, f(x)$$

- Variance

$$\mathrm{Var}[X] = \sum (x - \mathrm{E}[X])^2\, f(x)$$

- Mean

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance

$$\mathrm{Var}[X] = \int_{-\infty}^{\infty} \left(x - E(X)\right)^2 f(x) dx$$

Expectation of g(X)

$$\mathrm{E}[\mathrm{g}(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

- With the same probability, X takes a value within [a, b], where b>a.

Discrete version: toss a coin, roll a dice.

- What's the pdf?

- With the same probability, X takes a value within [a, b], where b>a.

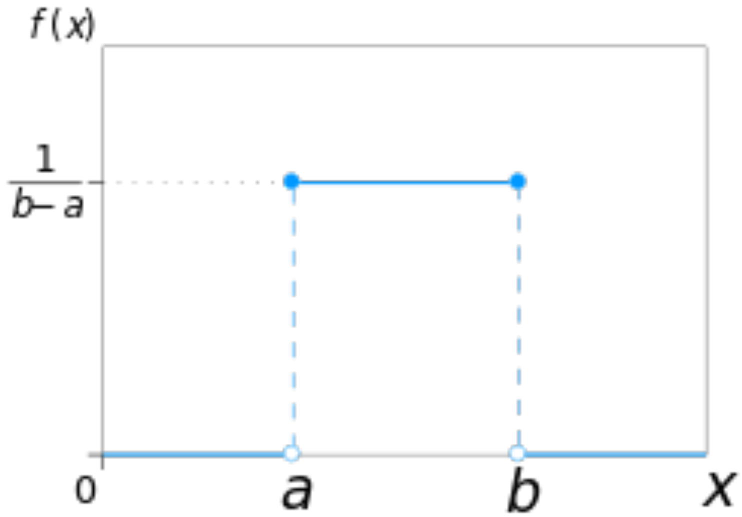
- What's the pdf?

- f(x) = c for $x \in [a, b]$ and f(x) = 0 for $x \notin [a, b]$
- As $\int_{-\infty}^{\infty} f(x)dx = c(b-a) = 1,$ we have

$$c = \frac{1}{b-a}$$

Uniform Distribution

- With the same probability, X takes a value within [a, b]
- $X{\sim}Uniform(a, b)$



Mean= $(a + b)/2$
Variance=$(b - a)^2/12$

Applications

- Given $X \sim Uniform(0,2)$
- What's the value of $E[2\, e^{X^2 + \cos(X)}]$?

Applications

- Given $X \sim Uniform(0,2)$
- What's the value of $E[2\, e^{X^2+\cos(X)}]$?

- f(x) = ½ for $x \in [0,2]$
- $E[2\, e^{X^2+\cos(X)}] = \int_0^2 2\, e^{x^2+\cos(x)} f(x)dx = \int_0^2 \; e^{x^2+\cos(x)}dx$

How to approximate $\int_0^2 e^{x^2+\cos(x)} dx$?

Given $X \sim Uniform(0,2)$, $E[2\, e^{X^2+\cos(X)}] = \int_0^2 e^{x^2+\cos(x)} dx$

- Draw N samples of $X \sim Uniform(0,2)$: $X_1, X_2, X_3, \ldots, X_N$
- Calculate $\frac{\sum_i 2\, e^{X_i^2+\cos(X_i)}}{N}$

Why? Expectation can be approximated by long-run average.

General Case

- How to calculate $\int_a^b h(x)dx$?

  - Draw N samples of $X \sim Uniform(a, b)$: $X_1, X_2, X_3, \ldots, X_N$
  - Calculate $\frac{\Sigma_i (\boldsymbol{b}-\boldsymbol{a})\, h(X_i)}{N}$

- $E[h(x)]$ only gives you the average "height" of $h(x)$
- In order to get $\int_a^b h(x)dx$, which is the area, we need to multiply $E[h(x)]$ by $(b - a)$

- Let $X \sim Uniform(a, b)$
- f(x) = 1/(b-a) for $x \in [a, b]$
- $E[(b-a)h(x)] = \int_a^b (b-a)h(x)f(x)dx = \int_a^b h(x)dx$

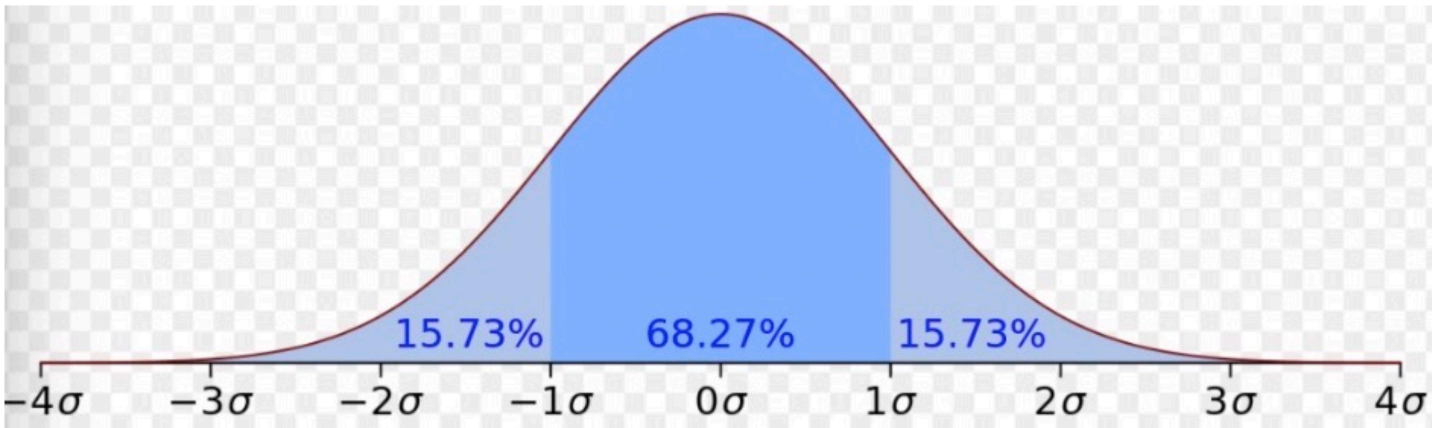
Most important:

Normal distribution

- X can be any real number
- Parameters: μ and σ

$$f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}\,\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)。$$

- $X \sim Normal(\mu, \sigma)$

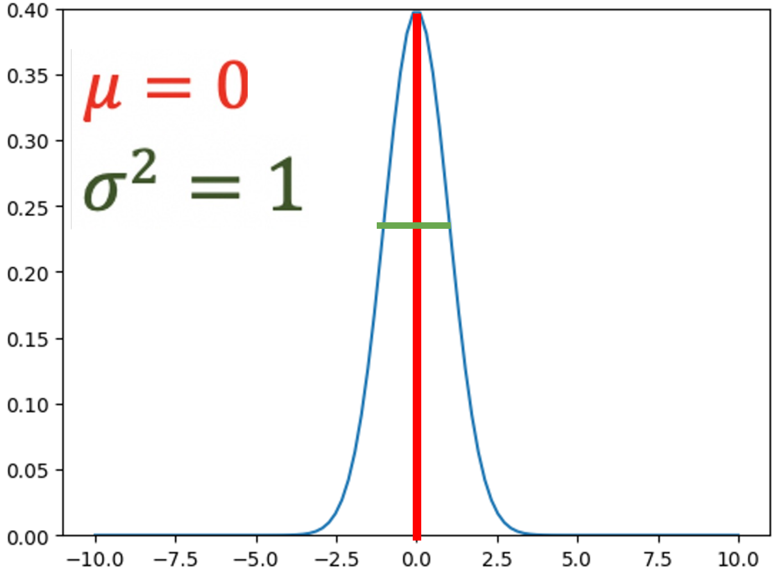


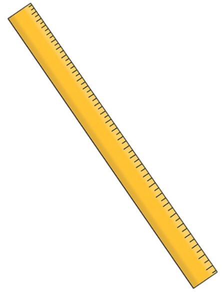
What is the meaning of the parameters?

Mean and Variance

- Mean: μ
- Variance: σ^2

The Normal PDF





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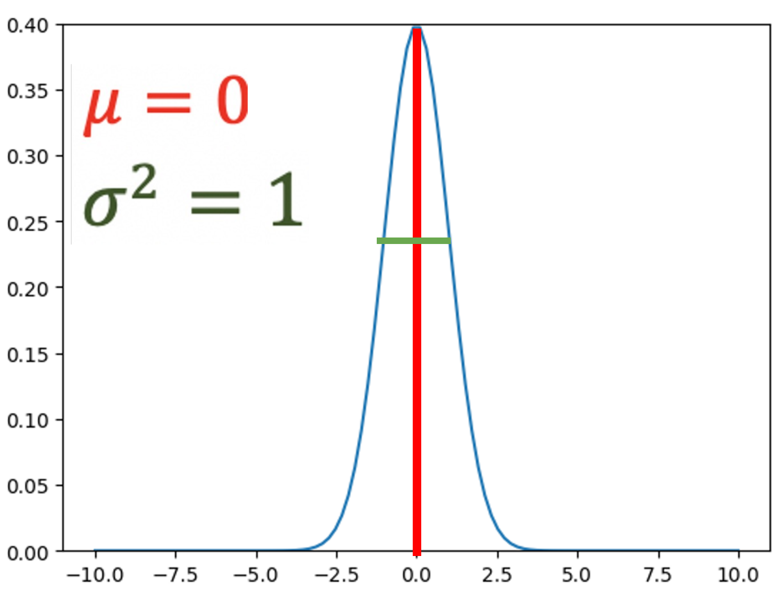

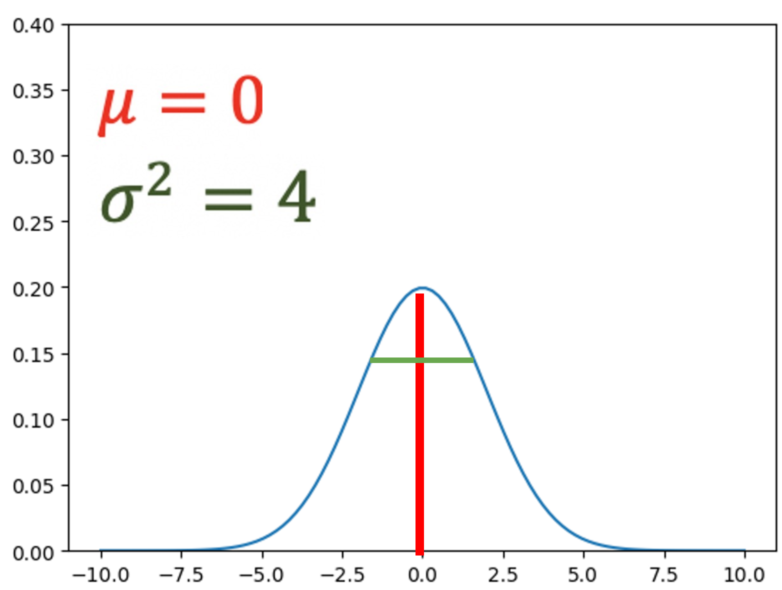

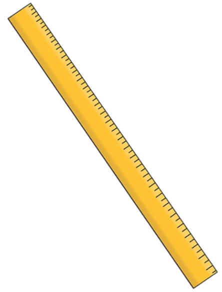

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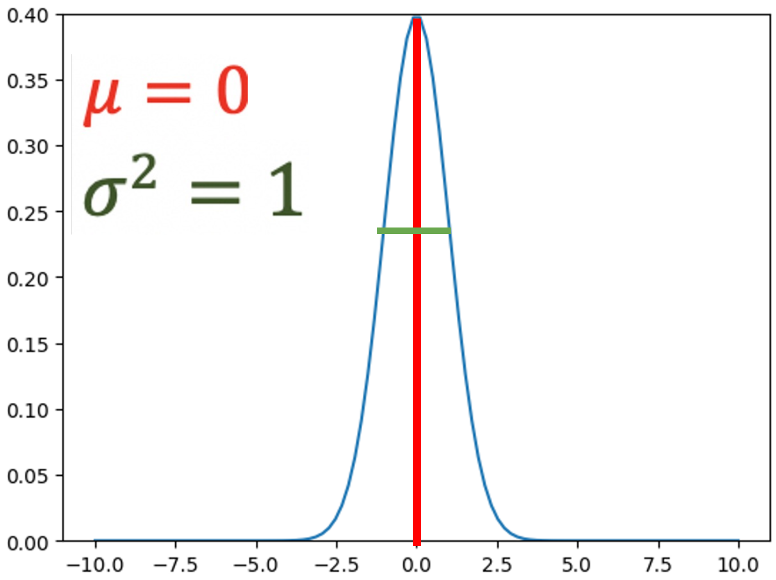

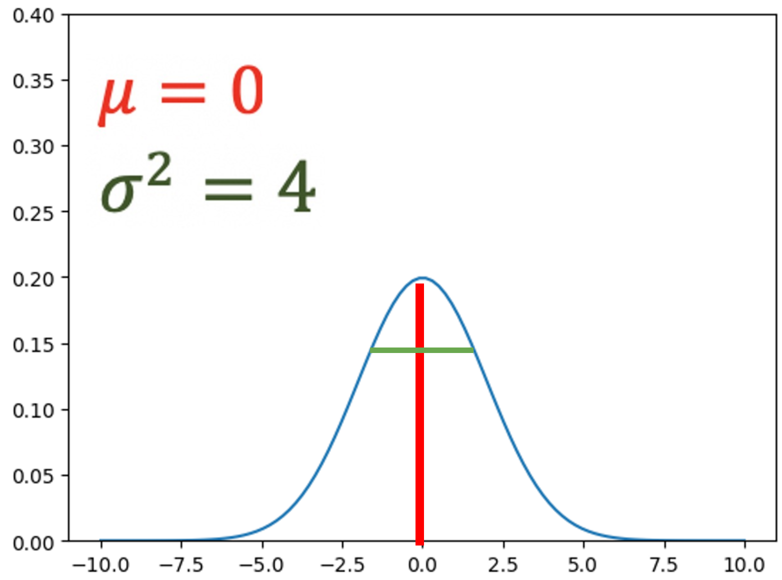

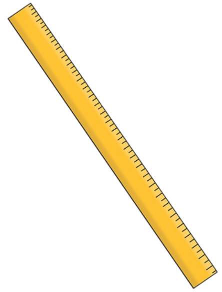

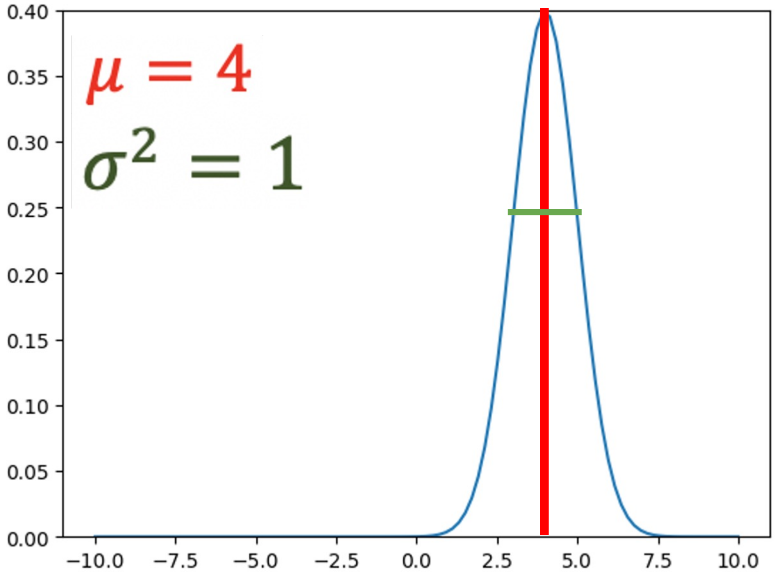

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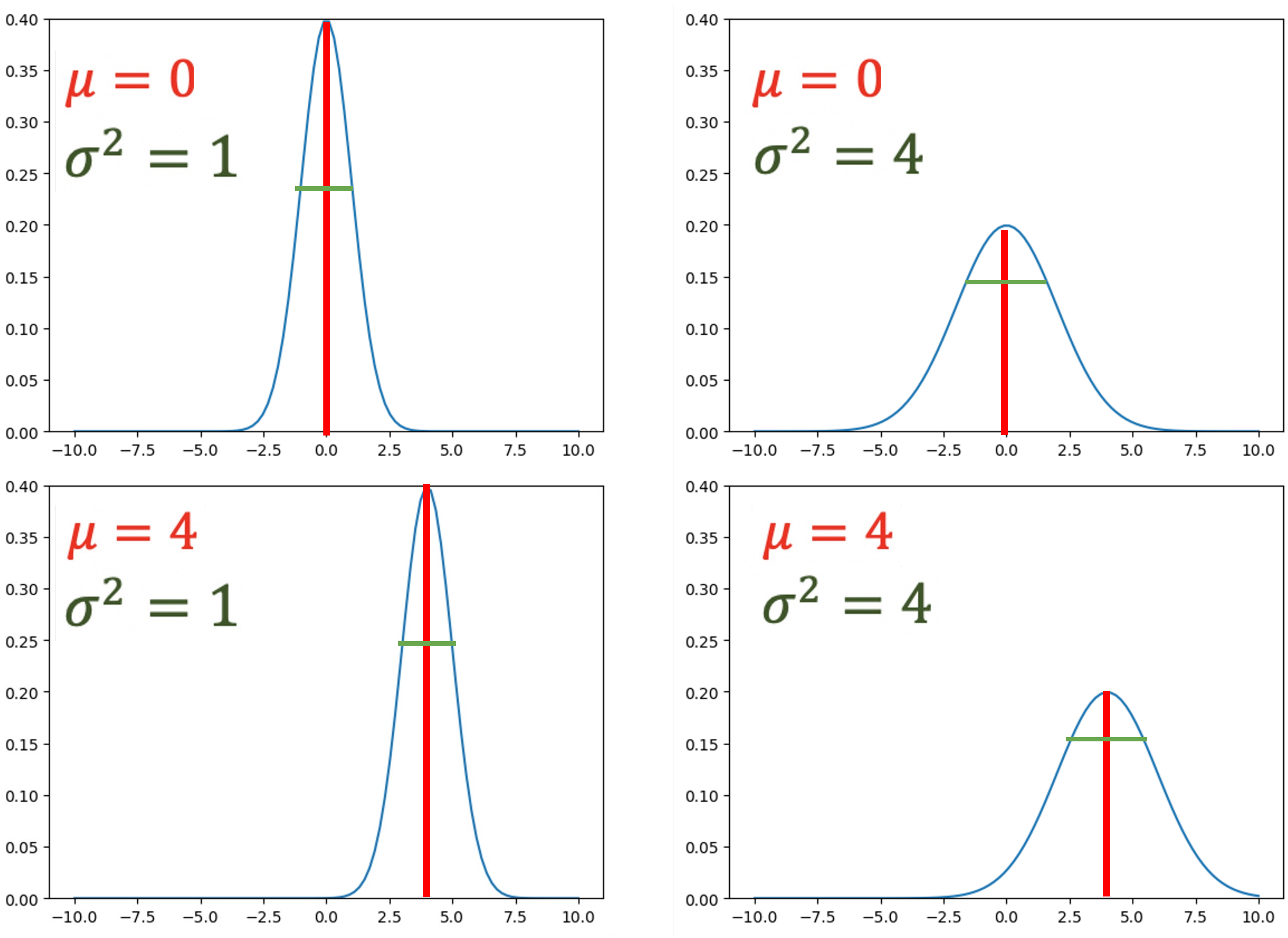
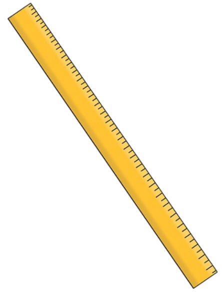




Why we have this distribution?

Normal Distribution

Central limit theorem

Lindeberg–Lévy CLT. Suppose $\{X_1, \ldots, X_n\}$ is a sequence of i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\operatorname{Var}[X_i] = \sigma^2 < \infty$. Then as n approaches infinity, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$:

$$\sqrt{n}\left(\bar{X}_n - \mu\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right).$$

no need to grasp!!!

- Let $X_1, X_2, X_3, \dots, X_n$ be the outcomes of n independent and identical trials of X.
- Define $\bar{X} = (X_1 + X_2 + X_3 + \cdots + X_n)/\mathrm{n}$
- Then, when n is large

$$\bar{X} \sim Normal(E[X], \frac{Var(X)}{n})$$

X may not be a normal distribution

X_i's are random

- Let $g(X_1), g(X_2), g(X_3), \dots, g(X_n)$ be the outcomes of n independent and identical trials of $\mathrm{g}(X)$.

- Define $\overline{g(X)} = (g(X_1) + g(X_2) + g(X_3), \dots + g(X_n))/\mathrm{n}$

- Then, when n is large

$$\overline{g(X)} \sim Normal(E[g(X)], \frac{Var(g(X))}{n})$$

Implication

- We do n independent and identical trials of $\mathrm{g}(X)$.
- Let our records be $g(x_1), g(x_2), g(x_3), \dots, g(x_n)$

- If n is large, very likely

$$\frac{g(x_1) + g(x_2) + g(x_3) + \cdots + g(x_n)}{n} \text{ is very close to } E[g(X)]$$

Review

- How to approximate $\int_a^b h(x)dx$?

- $X{\sim}Uniform(a, b)$
- $\int_a^b h(x)dx = E[(b-a)h(X)]$

(a common programming language has some built-in functions to do this)

- Draw n samples of $X{\sim}Uniform(a, b)$: x_1, x_2, x_3, …. , x_N
- Calculate $\frac{\Sigma_i (b-a)h(x_i)}{n}$

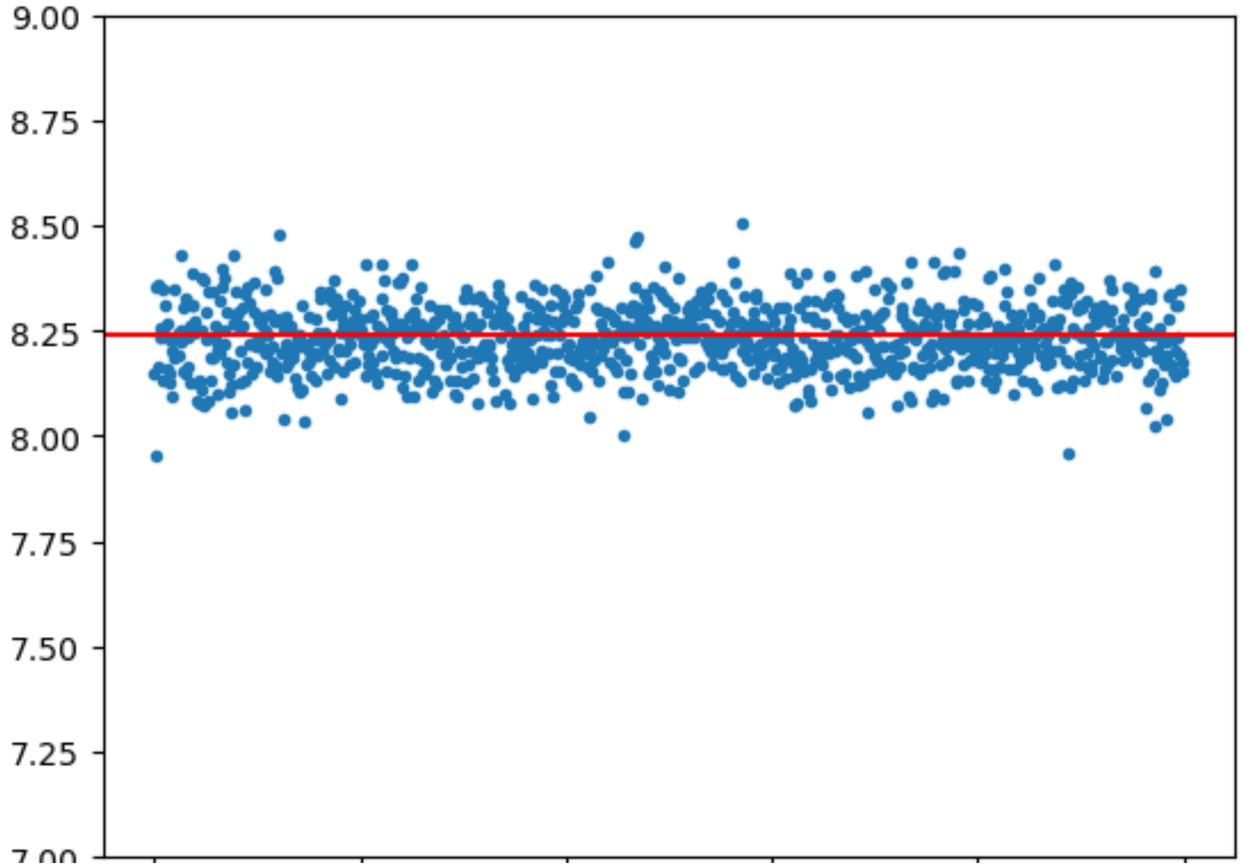
Result by discretization

- I divide the interval [0,2] into 10^8 sub-intervals with equal length.

- $\int_0^2 e^{x^2+\cos(x)}dx \approx$ 8.238253098458285

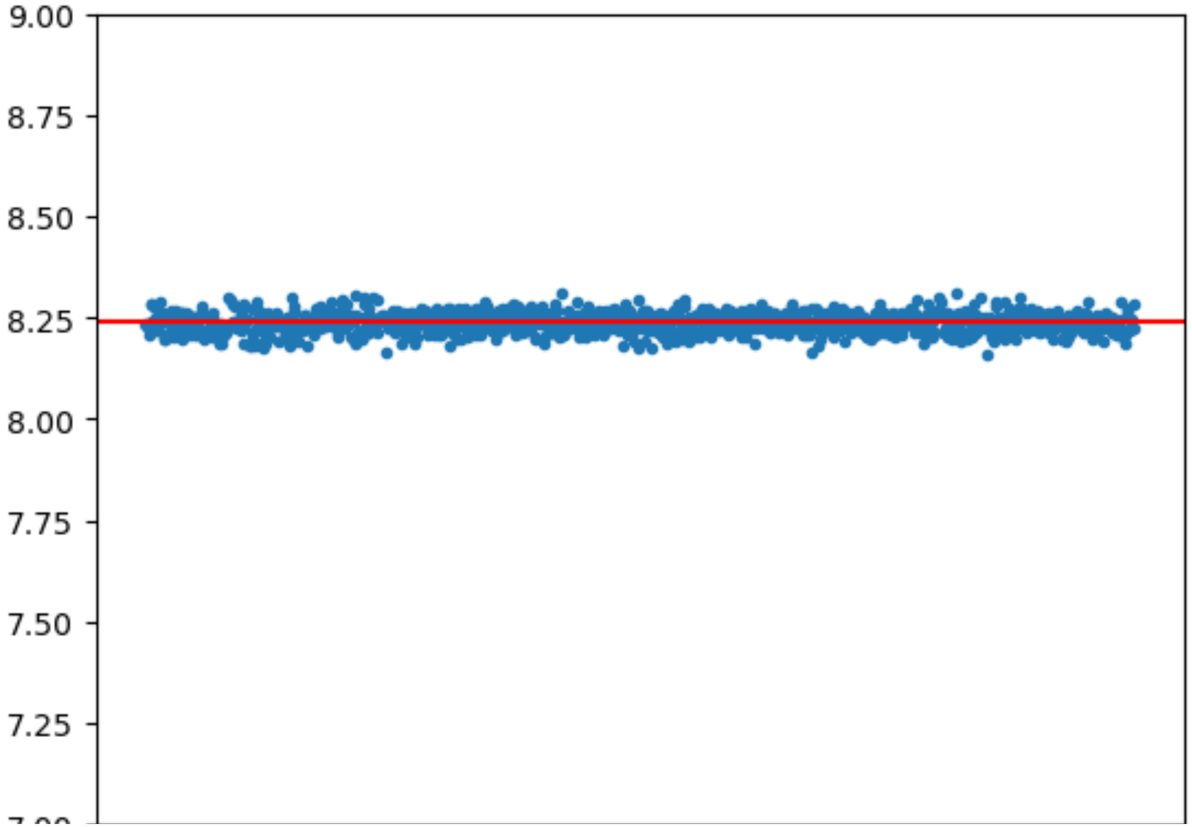
Approximation by sample mean

- 10^4 experiments



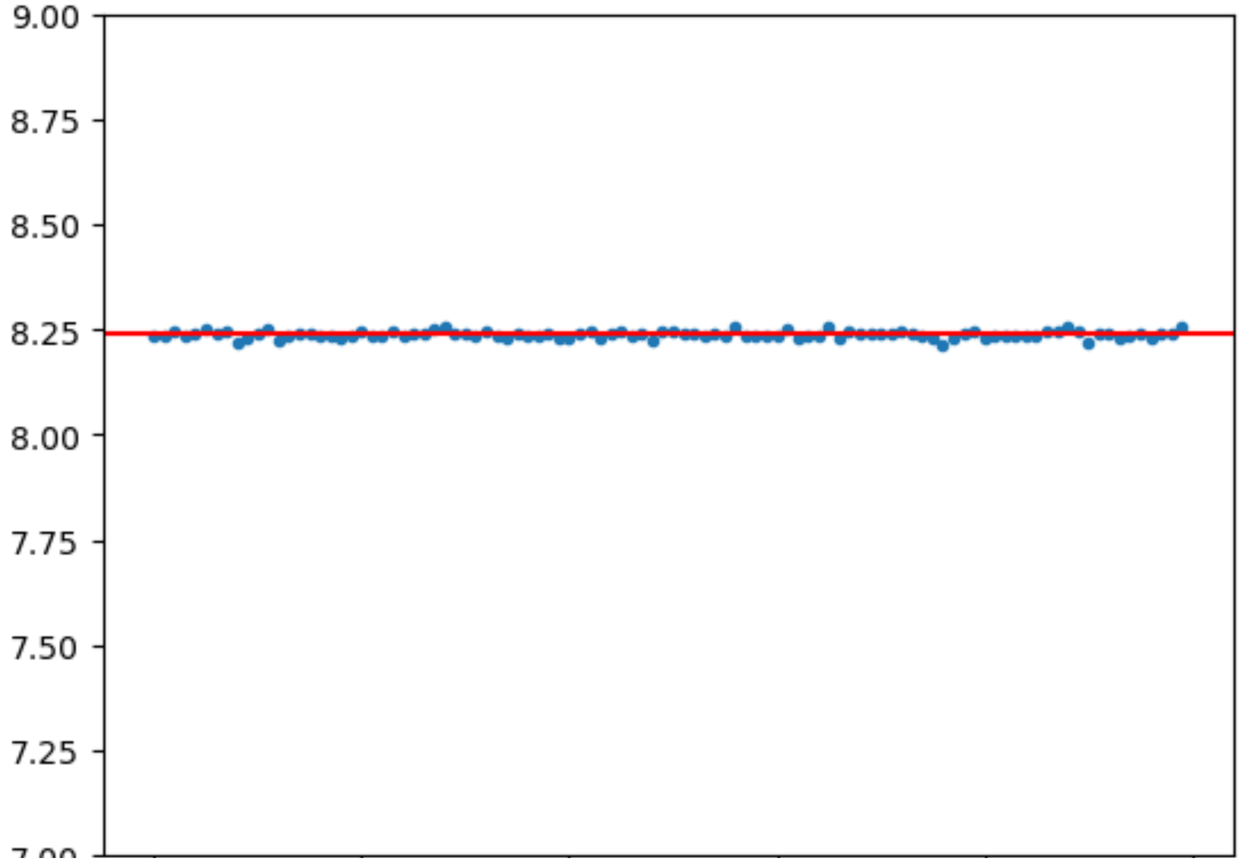
Approximation by sample mean

- 10^5 experiments



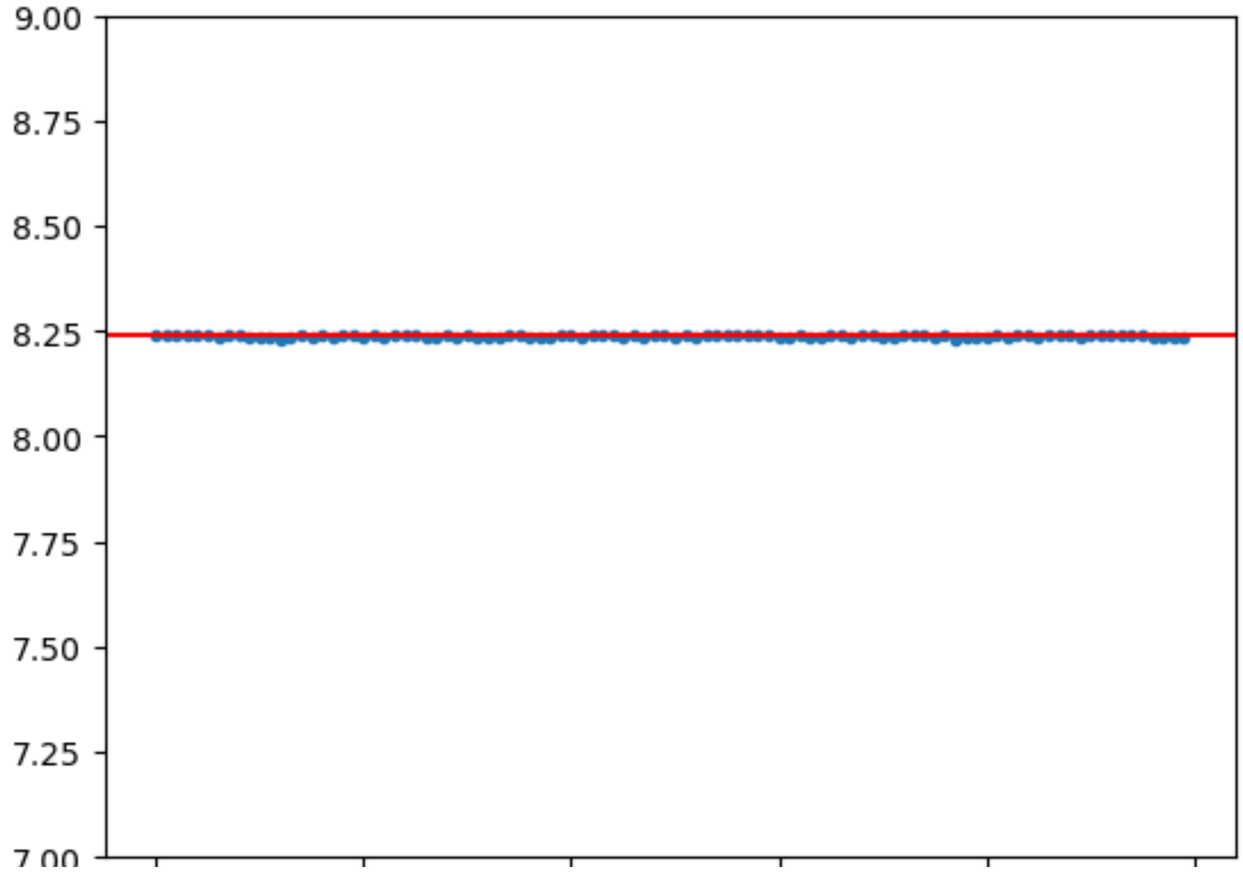
Approximation by sample mean

- 10^6 experiments



Approximation by sample mean

- 10^7 experiments



In our course (for one dimension)

- We will ask: How to approximate $\int_a^b h(x)dx$?

You answer:

- If $X \sim Uniform(a, b)$, then the pdf of X will be…
- I can prove that $E[(b-a)h(X)] = \ldots = \int_a^b h(x)dx$
- Thus, by **Lindeberg–Lévy CLT, I can approximate the integral by**

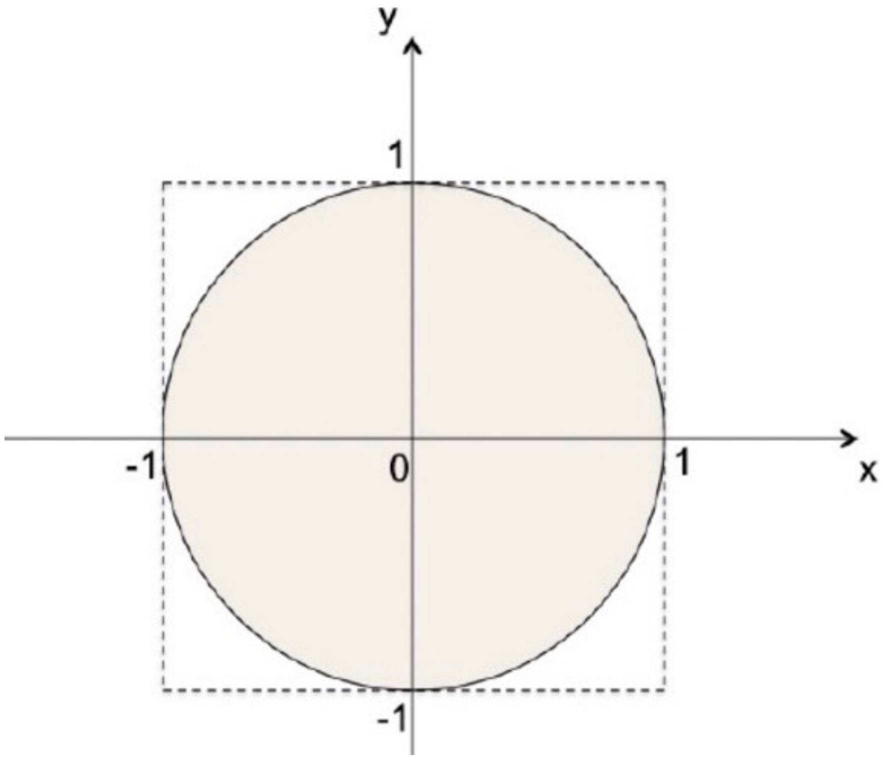
  - Draw n samples of $X \sim Uniform(a, b)$: $x_1, x_2, x_3, \ldots, x_N$
  - Calculate $\frac{\Sigma_i (b-a)h(x_i)}{n}$

Estimation of π

Draw a two-dimensional point from the square

$X \sim Uniform(-1,1)$

$Y \sim Uniform(-1,1)$



The sample proportion that $X^2 + Y^2 \leq 1$
is approximately $\pi/4$

In our course (for two dimensions)

- We will ask: How to approximate π?

- You answer:

Step 1:

- Draw a two-dimensional point from the square
  - $X \sim Uniform(-1,1)$
  - $\mathrm{Y} \sim Uniform(-1,1)$
- Then with the same chance (X,Y) is any point in the square

In our course (for two dimensions)

- We will ask: How to approximate π?

- You answer:

Step 2:
Let g(X,Y) = 1 if $X^2 + Y^2 \leq 1$ and 0 otherwise.
Then as area represents probability, E[g(X,Y)]= $\pi/4$

In our course (for two dimensions)

- We will ask: How to approximate π?

- You answer:

Step 3:

- Thus, by **Lindeberg–Lévy CLT, I can approximate π by**

  - Draw n samples of $X \sim Uniform(0,1)$: $x_1, x_2, x_3, \ldots, x_N$
  - Draw n samples of $Y \sim Uniform(0,1)$: $y_1, y_2, y_3, \ldots, y_N$
  - Calculate the proportion of $x_i^2 + y_i^2 \leq 1$

In our course (for two dimensions)

- We will ask: How to approximate π?

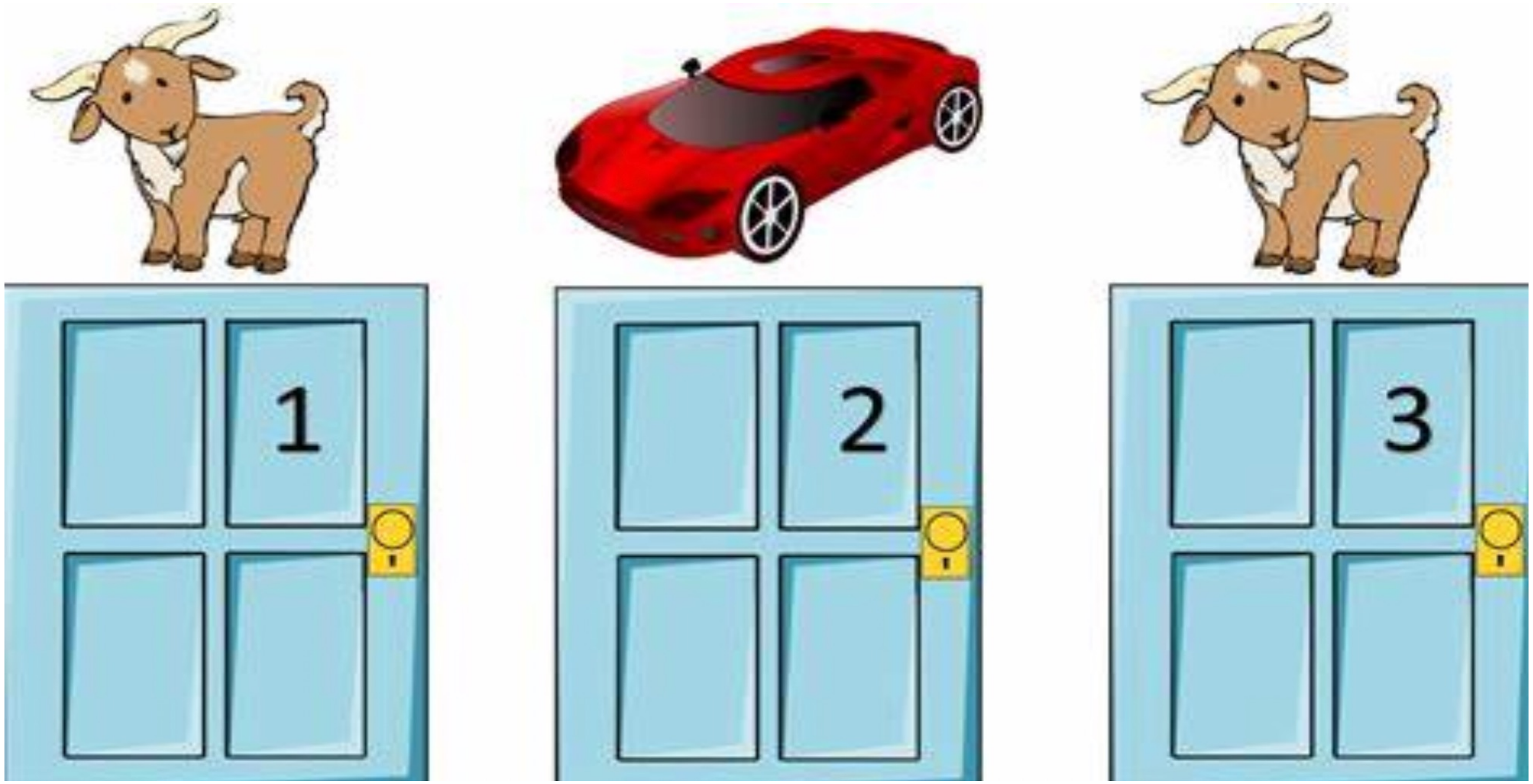
- You answer:

Step 4:

- Then π is approximated by the proportion calculated in step 3, multiplied by 4.

Advanced Concepts:
Correlation and Conditional Probability

Monty Hall problem



Monty Hall problem

Suppose you're on a game show, and you're given the choice of three doors:

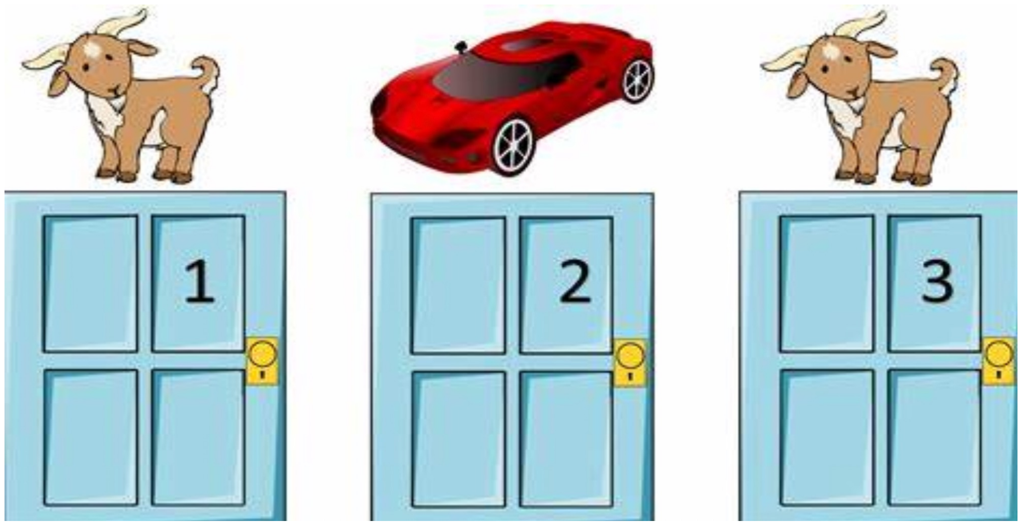
- Behind one door is a car;
- Behind the others, goats.

The car is randomly placed behind one door.

Which door you want to open?

Any will be OK

- After you choose No. 1 (without opening it), I open a door with a goat behind, do you want to change your door?
- Now you have additional information\event!
- Given the information, you may make a better decision!



Behind door 1	Behind door 2	Behind door 3	Result if staying at door #1	Result if switching to the door offered
Goat	Goat	**Car**	Wins goat	**Wins car**
Goat	**Car**	Goat	Wins goat	**Wins car**
Car	Goat	Goat	**Wins car**	Wins goat

P(win when not switch)=1/3

P(win when switch) = 2/3

If a car is worth 1 dollar and a goat 0 dollar, how much money will you at most pay to switch? 1/3

Correlation and conditional probability

Suggested reading material:

- Applied Statistics and Probability for Engineers, Third Edition, Douglas C. Montgomery and George C. Runger.
  - Chapter 5.1, 5.2

- Havard Stat 110, book by Joe Blitzstein and Jessica Hwang
  - https://drive.google.com/file/d/1VmkAAGOYCTORq1wxSQqy255qLJjTNvBl/edit Chapter 2.

- Given the realization of X, the distribution of Y may change (vice versa)

- We need a joint distribution to represent the probability of the outcome of the pair (X,Y)

Examples

- X and Y can take values in a set

- X: the temperature outside tomorrow
- Y: the number of households using air conditioners

- X: the age of a random selected person
- Y: the height of this person

- X: the attendance rate of a random selected student
- Y: the final score of this student

Joint Distribution (discrete case)

- f(x,y) is the probability such that X=x and Y=y.
- $\sum_{x,y} f(x,y) = 1$, $f(x,y) \in [0,1]$.
- $\sum_{y} f(x,y)$ is the probability that X=x
- $\sum_{x} f(x,y)$ is the probability that Y=y

The above f(x,y) is the p.m.f. for discrete case, and can be similarly defined as p.d.f. for the continuous case.

To represent the relationship

1. Correlation

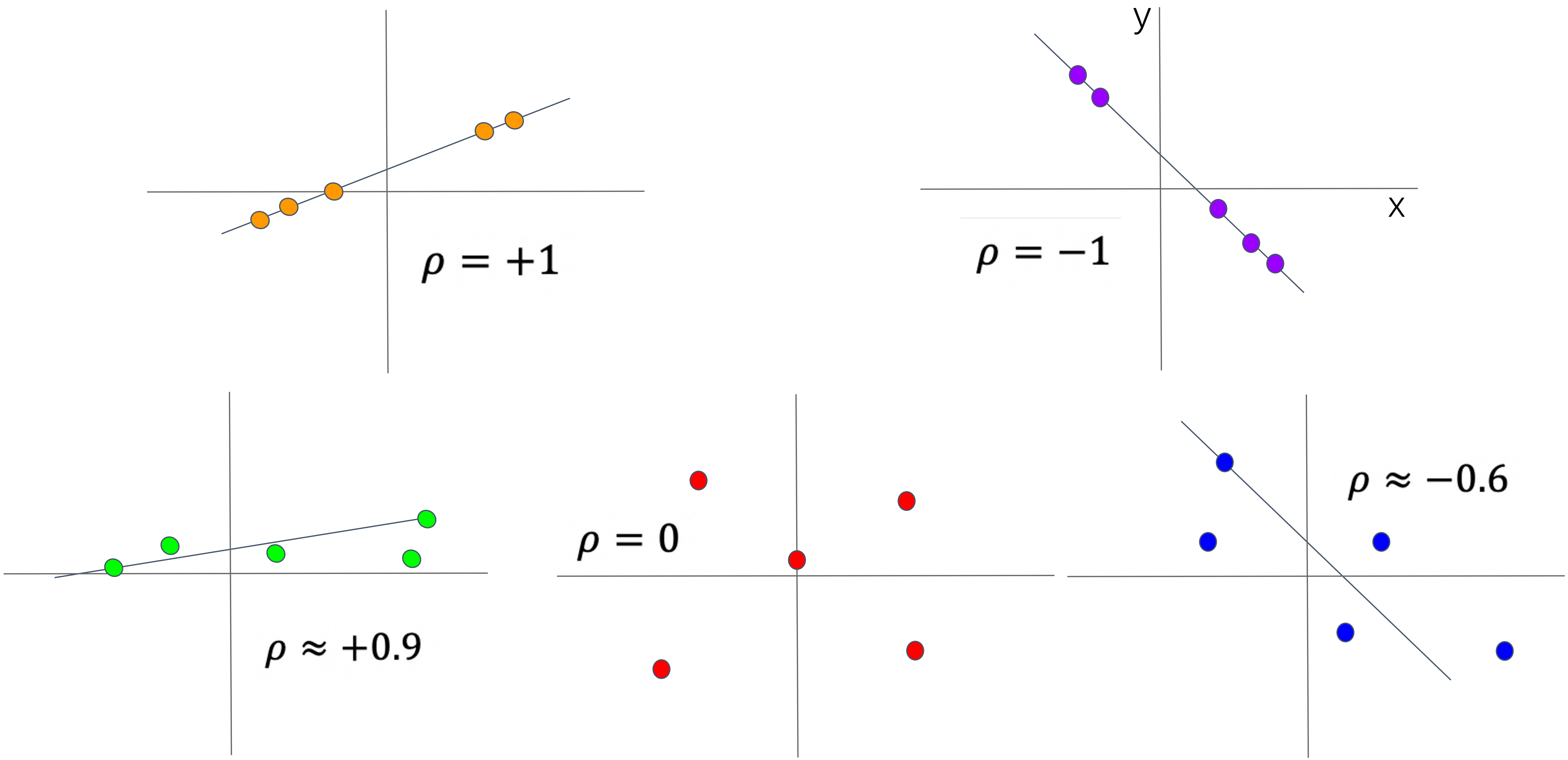
The value of correlation between two RVs is within [-1,1].
We often denote it by ρ

Graphical Illustration

- Let (X,Y) be a pair of two random variables.

- We draw all the possible outcomes of (X,Y) in a figure.

Graphical Illustration



Implication

- With a positive correlation (positively correlated),
  - Larger x implies larger y (vice versa)

- With zero correlation (uncorrelated),
  - No clear (linear) relationship between x and y

- With a negative correlation (negatively correlated),
  - Larger x implies smaller y (vice versa)

Mathematical Definition

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}}$$

- The numerator is called covariance
$$Cov(X,Y) = E[(X - E[X])(Y - E[Y])]$$

- The denominator is used to ensure $\rho \in [-1,1]$.

Now let's see how this quantity can give the implications we just discussed!

Covariance (Intuition)

$$Cov(X,Y) = E[(X - E[X])(Y - E[Y])]$$

$X > E[X], Y > E[Y] \quad \rightarrow \quad +$

$X < E[X], Y < E[Y] \quad \rightarrow \quad +$

$X < E[X], Y > E[Y] \quad \rightarrow \quad -$

$X > E[X], Y < E[Y] \quad \rightarrow \quad -$

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

Covariance (Intuition)

- Positive covariance:
  - The greater (lesser) values of one RV mainly correspond with the greater (lesser) values of the other RV.
  - that is, X and Y tend to show similar behavior.

- Negative covariance
  - The greater (lesser) values of one RV mainly correspond with the lesser (greater) values of the other RV.
  - that is, X and Y tend to show opposite behavior.

Remark

Understand why?

- $E[XY] = \sum_{x,y} f(x,y)xy$
- $E[X] = \sum_{x,y} f(x,y)x$
- $E[Y] = \sum_{x,y} f(x,y)y$

. $\mathbf{E}[XY] = \sum_{x,y} f(x,y)xy$

. $\mathbf{E}[X] = \sum_{x} \left(\sum_{y} f(x,y)\right) x$

. $\mathbf{E}[Y] = \sum_{y} (\sum_{x} f(x,y))\, y$

More about correlation

- X: amount of exercise
- Y: skin cancer patients

- A statistically significant positive correlation

- A conclusion: exercise can cause cancer to some extent!

True or false?

- Eat ice cream
- Use electric fan

- A statistically significant positive correlation

- A conclusion
  - Eat ice cream causes to use electric fan?
  - Use electric fan causes to eat ice cream?

Hot weather!!!

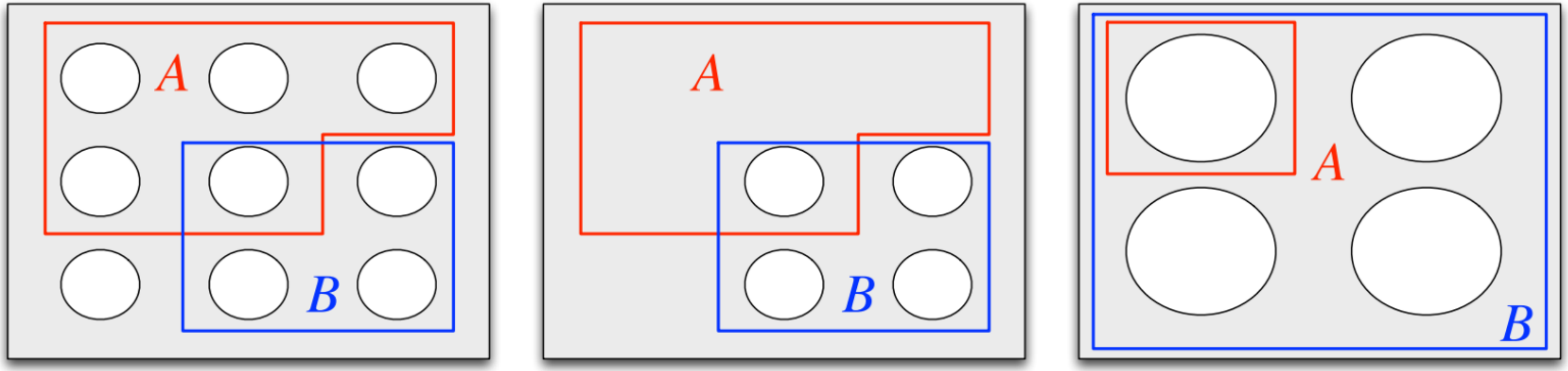
Correlation does not imply causality!

2. Conditional probability

- Given the realization of event A, the probability of event B may change

- (Conditional probability) If A and B are events with P(B)>0, then the conditional probability of A given B, denoted by P(A|B), is defined as $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



$P(A|B)=?$ $P(A|B)=?$ $P(A|B)=?$

- Given the realization of event A, the probability of event B may change

- (Conditional probability) If A and B are events with P(B)>0, then the conditional probability of A given B, denoted by P(A|B), is defined as $P(A|B) = \frac{P(A \cap B)}{P(B)}$

- (Independence) Two events A and B are called independent if and only if $P(A \cap B) = \mathrm{P}(A)P(B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \mathrm{P(A)}$$

- Given the realization of X, the distribution of Y may change

- Given X=x, the probability that Y=y.
- $f(Y=y|X=x)$ or $f(y|x)$ is **called** the conditional probability.
    - X and Y independent $\Leftrightarrow$ $f(y|x)=f(y)$ for all x,y
- Clearly, $f(x,y) = f(x)\, f(Y=y|X=x)$
- Thus, $f(Y=y|X=x) = f(x,y)/f(x)$ (**bayes rule**)

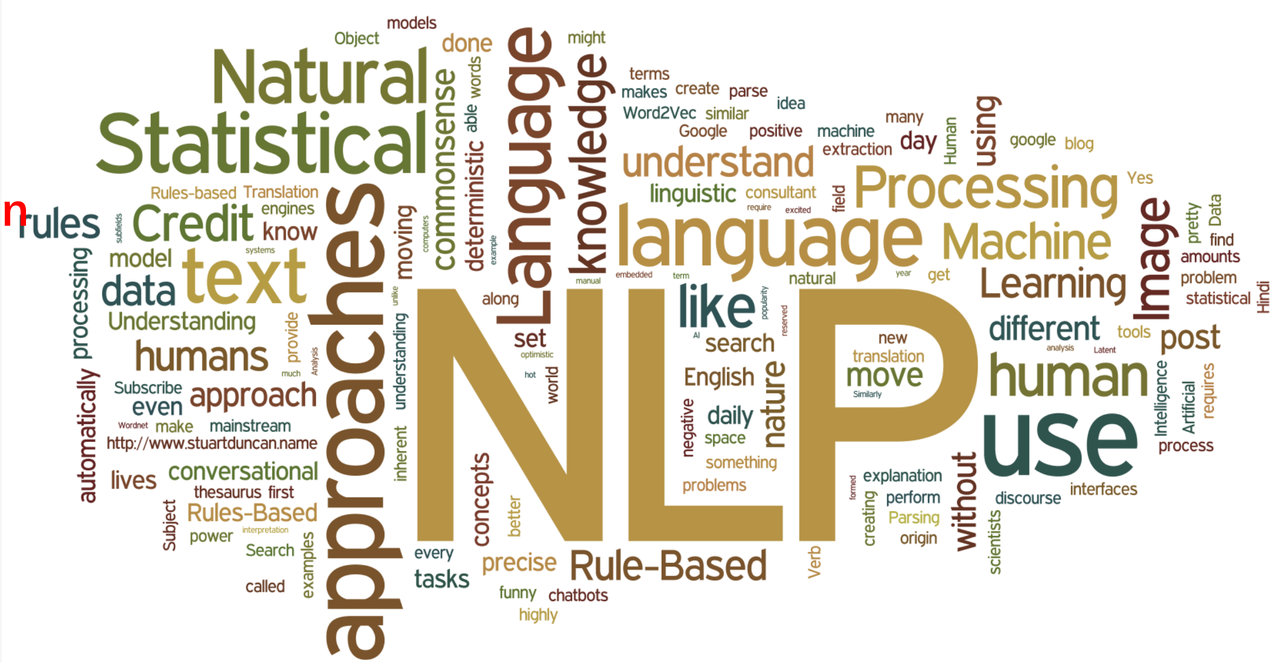
Conditional Probability in NLP

- H: mention “happy” in message
- D: mention “DDA2001” in message

- $P(H) = .1$
- $P(D) = .1$
- $P(H, D) = .08$

Known Information from data

- $P(H|D) = ??$



Conditional Probability in NLP

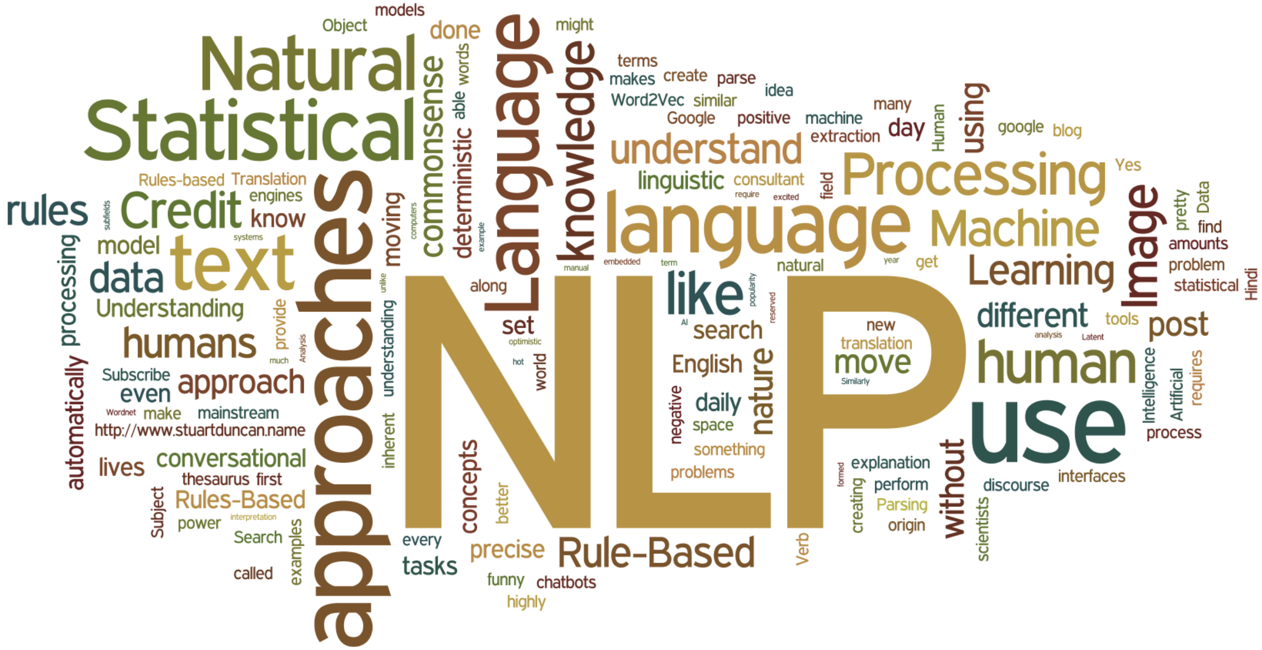
$f(y|x) = P(x,y) / f(x)$

- H: mention "happy" in message
- D: mention "DDA2001" in message

- $P(H) = .1$
- $P(D) = .1$
- $P(H, D) = .08$

- $P(H|D)$

  $= 0.08/0.1 = 0.8$



Conditional Probability in Revenue Management

- A (B,C) : buy product A (B,C)
- Earn the same from selling any product
- P(A and B) = .1   P(A and C) =.9

$f(y|x) = f(x,y) / f(x)$

- If a consumer buys A first, which product shall be recommended next?



Conditional Probability in Real life

- Why do we need to register our gender when using Taobao?

- Automatic spelling correction

- Nucleic acid test

- Weather forecast

- Do you have other examples?

More about conditional probability

- Will $P(B|A)$ and $P(A|B)$ be significantly different?

- COVID-19: infected or well
- Nucleic acid test: positive or negative

- $P(\text{positive} \mid \text{Infected}) = 0.99$
- $P(\text{infected} \mid \text{positive})$?

$P(\text{Infected}) = 0.01$

$P(\text{well}) = 0.99$

$P(\text{positive} \mid \text{Infected}) = 0.99,$

$P(\text{negative} \mid \text{Infected}) = 0.01,$

$P(\text{positive} \mid \text{well}) = 0.01,$

$P(\text{negative} \mid \text{well}) = 0.99,$

Infected and Positive 0.0099	Infected and negative 0.0001
Well and Positive	Well and Negative

P(Infected) = 0.01
P(well) = 0.99
P(positive | Infected) = 0.99,
P(negative | Infected) = 0.01,
P(positive | well) = 0.01,
P(negative | well) = 0.99,

$$P(A \text{ and } B) = P(A) \times P(B|A)$$

P(Infected and positive) = 0.0099
P(Infected and negative) = 0.0001

Infected and Positive 0.0099	Infected and negative 0.0001
Well and Positive 0.0099	Well and Negative 0.9801

P(Infected) = 0.01
P(well) = 0.99
P(positive | Infected) = 0.99,
P(negative | Infected) = 0.01,
P(positive | well) = 0.01,
P(negative | well) = 0.99,

$$P(A \text{ and } B) = P(A) \times P(B|A)$$

P(well and positive) = 0.0099
P(well and negative) = 0.9801

Infected and Positive 0.0099	Infected and negative 0.0001
Well and Positive 0.0099	Well and Negative 0.9801

P(Infected) = 0.01
P(well) = 0.99
P(positive | Infected) = 0.99,
P(negative | Infected) = 0.01,
P(positive | well) = 0.01,
P(negative | well) = 0.99,

Different!

P(A| B) = P(*B* and A)/P(B)

P(infected | positive) = 0.50

Remark

- We always have P(negative | Infected) + P(positive | Infected) =1. This is because conditional on "infected", the outcome is always either negative or positive. Similarly for P(negative | well) + P(positive | well) =1.

- But there is no guarantee on the value of P(A | Infected) + P(B | well) for any A and B. This is because the probability is defined conditional on different events.

- More applications:

- In medical diagnosis, conditional probability is often used to assess the probability of a particular disease given certain symptoms or test results. For instance, the probability of having a specific type of cancer given the results of a diagnostic test.

- More applications:

- Conditional probability is applied in finance for risk management and portfolio optimization. Investors use conditional probabilities to estimate the likelihood of different market conditions based on historical data and economic indicators.

- More applications:

- Meteorologists use conditional probability to predict the likelihood of rain, snow, or other weather events based on current atmospheric conditions. For example, the probability of rain tomorrow given that it is already raining today.

- More applications:

- In the insurance industry, conditional probability is used to assess the likelihood of specific events occurring, such as car accidents or health issues, based on various factors like age, gender, and lifestyle.

- More applications:

- Manufacturing industries employ conditional probability to assess the probability of a defective product given certain production conditions or quality control parameters.

- More applications:

- Conditional probability is applied in game theory to model strategic interactions and decision-making. Players may update their strategies based on the conditional probabilities of opponents' moves.

- More applications:

- Ask chatgpt